

Amanuel Fesseha

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EDUCATION

University of Georgia

Bachelor of Science, Electrical Engineering

Athens, GA

May 2027

Georgia State University

Associates of Science, Electrical Engineering

Atlanta, GA

May 2024

Cumulative GPA: 3.5/4.0

TECHNICAL SKILLS

- **Software:** Altium Designer, KiCad (schematic/PCB), Autodesk (Fusion 360, Inventor, AutoCAD), MatLab
- **Robotics and ML:** IssacSim, ROS2, Gazebo, URDF/Xacro; PyTorch CNNs and transfer learning
- **Languages:** Python, C/C++, SQL; Tools: Git/GitHub, Jupyter: Java, Python, C
- **Embedded:** Design with ESP32, STM32, nRF52; Sensor integration and IO with I2C, SPI, UART

RELEVANT PROJECT EXPERIENCE

CURO Research Fellowship, *Biomedically Adaptive Leaping System*

Summer 2025 - 2026

- Engineered a bio-inspired leaping robot in Fusion 360 with focus on energy efficiency and actuator performance.
- Built custom brushless motor actuators with FOC control, delivering ~18 Nm torque with a quarter of the cost.
- Constructed URDF/Xacro for ROS2 Jazzy and Gazebo simulation, resolving mesh and transform issues.
- Optimized gear ratios and control pipelines to improve locomotion adaptability.

Electronic Testbench, *ECSE 2920 Design Methodology*

Spring 2026

- Designed a Pi Hat integrating Pi interfaces, analog circuitry, waveform generation, and power distribution.
- Routed a 4-layer mixed-signal PCB in Altium Designer, optimizing for manufacturability and system integration.
- Applied design rules and supported PCB fabrication, assembly, and hardware integration.

Sensor-Integrated Gimbal Headlight Tracker, *ELEE 4230 Sensors and Transducers*

Summer 2026

- Designed an adaptive bicycle headlight position tracking, ambient-light sensing, and two-axis servo control.
- Developed the signal-processing pipeline for sensor data, including hardware linearization, analog filtering, ADC conversion, calibration, averaging, and Kalman filtering for stable real-time control.

RELEVANT WORK EXPERIENCE

Create Lab

Athens, GA

Student Engineer

September 2024 - Present

- Engineered custom products with cross-functional teams, translating client briefs into manufacturable designs.
- Managed project timelines, ensuring efficient delivery of high-quality products within specified deadlines.
- Conducted product testing and provided technical feedback to optimize performance and functionality.
- Worked closely with clients to refine project requirements and align final product designs with their expectations.

Projects

- **Reaction Latency Evaluation** - Developed a modern Reaction Latency System using ESP32 and industrial-grade LED strips to deliver precise, reliable timing measurements for gerontology research.
- **Pecan Impact Recording Device** – Designed and Built a pecan-sized wireless impact-sensing PCB around nRF52840 MCU with a 6-axis IMU, implementing a 2.4 GHz antenna and matching network for wireless communication, and in high-rate sampling and pre/post-impact buffering with onboard logging.

INVOLVEMENT EXPERIENCE

UGA Robotics Club, *Member* | Athens, GA

August 2024 - Present

- Designed and built small-scale combat robots, integrating mechanical and electrical systems with lightweight components fabricated using additive manufacturing techniques. Conducted iterative testing and optimization to enhance performance and durability, utilizing engineering tools to get a competitive edge.